

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Lines, angles, and triangles

02

10 answers · Rationale-rich · Teach from doc

Q1**C**

C. The value of x is equal to 145.

Choice C is correct. Vertical angles, or angles that are opposite each other when two lines intersect, are congruent. It's given that line k intersects line n . Based on the figure, the angle with measure x° and the angle with measure 145° are vertical angles. Therefore, the value of x is equal to 145.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D**

D. 170

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 170° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 170^\circ$. Therefore, the value of w is 170.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3**D**

D. 40

Choice D is correct. Since $\angle BCD$ and $\angle BCA$ form a linear pair of angles, their measures sum to 180° . It's given that the measure of $\angle BCD$ is 110° . Therefore, $110^\circ + \angle BCA = 180^\circ$.

Subtracting 110° from both sides of this equation gives the measure of $\angle BCA$ as 70° . It's also given that the measure of $\angle BAC$ is equal to the measure of $\angle BCA$. Thus, the measure of $\angle BAC$ is also 70° . The measures of the interior angles of a triangle sum to 180° . Thus, $70^\circ + 70^\circ + x^\circ = 180^\circ$. Combining like terms on the left-hand side of this equation yields $140^\circ + x^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $x^\circ = 40^\circ$, or $x = 40$.

Choice A is incorrect. This is the value of the measure of $\angle BCD$. Choice B is incorrect. This is the value of the measure of each of the other two interior angles, $\angle BCA$ and $\angle BAC$. Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.

Q4**162**

The correct answer is 162. It's given that line r is parallel to line s . Since line n intersects both lines r and s , it's a transversal. The angles in the figure marked as 162° and x° are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is 162.

Q5**D****D.** 136

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**B****B.** 77

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of 90° . It's shown that the other two interior angles measure 13° and a° . The sum of the measures of the interior angles of a triangle is 180° ; therefore, $90 + 13 + a = 180$. Combining like terms on the left-hand side of this equation yields $103 + a = 180$. Subtracting 103 from both sides of this equation yields $a = 77$.

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of a .

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of a .

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of a .

Q7**D****D.** 24

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24.

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

Q8**A****A.** $AC = 4,680$ mm

Choice A is correct. In an equilateral triangle, all three sides have the same length. It's given that in triangle ABC , $AB = 4,680$ mm and $BC = 4,680$ mm. Therefore, if $AC = 4,680$ mm, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680$ mm is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**A****A.** 22

Choice A is correct. It's given that BC is parallel to AD . Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is 22.

Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of 22, not the value of y .

Choice D is incorrect. This is the supplement of 22, not the value of y .

Q10**B****B.** 45°

Choice B is correct. It's given that triangles EFG and JKL are congruent such that angle E corresponds to angle J . Corresponding angles of congruent triangles are congruent, so angle E and angle J must be congruent. Therefore, if the measure of angle E is 45° , then the measure of angle J is also 45° .

Choice A is incorrect. This is the measure of angle K , not angle J .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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